Self-Supervised Learning With Learnable Sparse Contrastive Sampling for Hyperspectral Image Classification

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***Abstract*— Contrastive learning (CL) with learnable examples performs outstandingly in data representation. However, when dealing with hard samples, instance-level alignment with excessive uniformity may descend into trivial clusters, especially when confronted with interclass similarity and intraclass diversity in hyperspectral images (HSIs). To solve this problem, we regard prototypical CL as tracing the potential probability density distribution. Then, a novel pretraining method, learnable sparse contrastive sampling (LSCoSa), is proposed for discriminative representation learning, containing sparse positive sampling and multiple positives learning. Specifically, on the basis of cooperative–adversarial CL, we first exert a Kullback–Leibler (KL) divergence regularizer on the average activation probability of the prototypes, suppressing fake density prototypes for sparse positive sampling. Furthermore, we propose multiple positives learning, in which the top-*k* potential positives are retrieved and dynamically weighted for contrastive supervision, to avoid trivial clusters and cover satisfying semantic variations. Comprehensive experiments on three HSI benchmark datasets demonstrate that LSCoSa achieves significant advantages over other HSI classification (HSIC) methods. The code is available at https://github.com/sakurashine/LSCoSa.**

***Index Terms*— Contrastive self-supervised learning (SSL), hyperspectral image (HSI), Kullback–Leibler (KL) divergence, positive and negative samples.**

I. Introduction

HYPERSPECTRAL imagery, developing significantly in spectral breadth and resolution, offers extensive cues to fine-ground object recognition in scenes for civilian and military applications. Meanwhile, it poses significant challenges for the distribution prediction of the data because of extreme spectral variation caused by environmental factors. Therefore, discriminative semantic representation and classification of hyperspectral images (HSIs) have always been a vibrant topic in efficiently utilizing data.

Deep neural networks exhibit excellent data perception power and are generally preferred in HSI classification (HSIC) [1]. However, end-to-end supervised learning usually depends upon manual labels to pull the same categories nearby and push different types apart. At this point, we typically need massive labeled samples to optimize the deep model. To address the overfitting from small set sample supervision, lightweight modeling [2], active learning [3], supervised contrastive learning (CL) [4], prototypical contrastive supervision refined few-shot learning [5], and other methods have been proposed for HSIC [6]. Even so, supervision with extremely sparse labels always converges to a weak solution and needs to improve generalization. Unlabeled raw data are more available than labeled samples and can provide rich information for semantic perception. Thus, self-supervised pretraining modeling receives extensive attention due to the superiority of mining structural information in data and the generalization to downstream tasks.

Self-supervised learning (SSL) expects model supervision derived from raw data themselves and learns representation by masked image modeling (MIM) [7], [8], CL [9], [10], [11], or other pretext tasks [12]. The pretrained model is then transferred to downstream tasks, which gains popularity in HSIC with a small set of samples. Autoencoder (AE) is initially employed for SSL of HSI, such as sparse AE [13] and superpixel-based relational AE [14]. As Transformer outstanding in computer vision, MIM is recognized as efficient in semantic representation of HSI [15], [16]. Compared with generative SSL, CL in latent embedding space is a family of the most potent methods, which confronts feature extraction head-on with fewer parameters, and is competitive with supervised learning, even under the linear evaluation protocol. This significantly boosts accurate HSI perception and classification, especially for HSI with subtle geometry distribution and minimal annotated labels. For example, a self-supervised CL model (SSCL) [17], a SSL model combining AE and prototypical CL (ContrastNet) [18], a Bootstrap Your Own Latent (BYOL)-based model with

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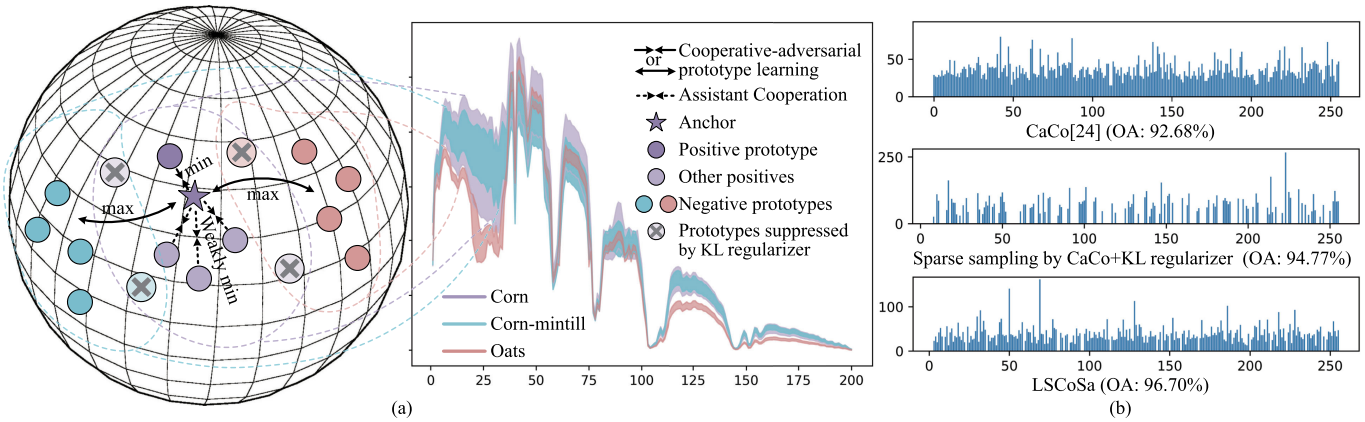


Fig. 1. Contrastive loss can result in fake density clusters and excessive uniformity during dynamic dictionary learning, especially when many different categories mix up in a similar spectrum. It is necessary to suppress prototypes uncertain as positives and encourage nontrivial clusters for more semantic variations, as illustrated in (a). Thus, we employ KL divergence regularization and multiple positives learning for sparse sampling. (b) Chosen counts of the prototype as positive samples with different sampling strategies. The results verify our hypothesis. Take the IP data, for instance.

Transformer backbone (BYOL-T) [19], and other self-supervised models [20], [21], [22], [23] all build contrastive supervision frameworks for HSIC and achieve approving performance.

Contrastive SSL trains deep models by aligning anchors with similar positive samples and distinguishing them from dissimilar negatives. We can regard it as anchor querying in a dynamic dictionary [10], where the contrastive counterparts reside. The existing methods construct this dictionary through heuristic collections or trainable parameters, with a shared motivation: aligning positives to ensure instance traceability and maintaining uniform representations to prevent encoder collapse. However, several dilemmas exist in terms of adequate supervision. These dilemmas include the trade-off between instance-level alignment and intraclass similarity and diversity, balancing the need for rich enough negatives with concerns about memory overhead and lag during updating, and the challenge of pursuing hard negatives versus misjudging possible positives. A cooperative-adversarial contrastive model (CaCo) [24] performs CL with learnable examples, which specializes in tracing hard negatives and is device friendly. However, it biases excessively toward uniformity on the unit hypersphere, implying that possible positives are pushed apart indiscriminately as negatives. This is undesirable for discriminative feature embedding in HSIC.

When we regard the dictionary as a memory bank for storing prototypes of clusters, prototypical CL can be treated as tracing the potential probability density distribution, with clusters representing local maxima in this density. HSI exhibits severe spectral uncertainty, where many different categories mix up in a similar spectrum with substantial intraclass diversity, as the spectral reflectance curve drawn in the middle of Fig. 1. In this scenario, the local density modes become much more challenging to trace. Instance-level contrastive discrimination can, thus, effortlessly encourage fake density clustering and excessive uniformity, which deviates from our intended divergence.

In this article, we wish to suppress sampling the prototypes as positives when they are retrieved with a low probability, as the suppressed prototypes illustrated in Fig. 1(a). To achieve suppression, we impose sparsity constraints, such as Kullback–Leibler (KL) divergence regularizer, on the average chosen probability of the prototypes, forcing them to be activated sparsely. Fig. 1(b) shows the frequency distribution of all the prototypes chosen as positives. The KL regularizer restricts the positives to be sampled from only a fraction of prototypes but not spread to cover the whole dictionary. With linear classifier evaluation, sparse sampling with KL regularizer achieves a more than 2% improvement over CaCo [24].

The above conclusion implies that we select all prototypes in the dictionary to serve as negative counterparts while only a portion as positives, which appears to be a cognitive decision. Fewer positives prevent the prototypes from focusing much on regions of low density, while enough negative samples can cover rich semantics and avoid model collapse. However, single instance positives alignment may descend into trivial clusters, which cannot cover all the semantic variances within a given category [25], let alone considering the well-known intraclass diversity in HSI. Thus, combined with the preceding KL regularizer, we propose an SSL method with learnable sparse contrastive sampling (LSCoSa). We retrieve the top-k potential positives and then adaptively weigh the last $k-1$ positive samples to weakly minimize their distance with the anchor and assist the most one in achieving generalization alignment [as illustrated in Fig. 1(a)], encouraging clusters with more semantic variations. The average chosen of prototypes and linear classifier evaluation further verifies our hypothesis. LSCoSa outperforms CaCo [24] from 92.68% to 96.70% on Indian Pines (IP) data classification.

In summary, we make the following contributions.

1) We propose an SSL method of ResNet backbone and LSCoSa for HSIC, which excels at representation learning of datasets in severe spectral uncertainty, such as intraclass diversity and interclass similarity. Comprehensive experiments on three HSI datasets demonstrate that LSCoSa reaches the anticipated effects over benchmark CaCo and is superior to other SSL models under linear probing with a few training samples.
2) To prevent clustering at noise-induced low-density regions when pursuing uniform representation in contrastive SSL with learnable prototypes, we exert KL divergence regularizer on average activate probability

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of the prototypes to achieve sparse positive sampling, which focuses on tracing regions with local maxima and discarding the fake clusters. The positive probability visualization and linear probing results ideally confirm our assumption and expectation.

3) To alleviate trivial clustering in contrastive supervision with single instance positives, we retrieve the top-k probable positives, the most similar one for alignment to the anchor, and the others assigned with dynamic weights to keep cluster diversity that covers more semantic variations for generalization representation. Ablation experiments show that multiple positives learning achieves remarkable performance in linear classification, alone or especially in partnership with the KL regularizer.

The remainder of this article is organized as follows. In Section II, we will introduce the related works about sampling efficiency, and in Section III, we will detail the proposed LSCoSa method. A comprehensive ablation study and comparative experiments will be presented in Section IV to demonstrate the validity of this method. This article will conclude in Section V.

II. Related Works

Contrastive SSL learns data representation by drawing positive samples together and pushing apart negatives. This section will review related works on sample selection for CL and then pose challenges toward HSIC.

A. Sample Selection for Contrastive Learning

In CL, choosing influential contrastive samples is crucial for model optimization without manual label guidance [25]. Most recent works have validated that data augmentation and other multiple views from the same input are effective and crucial in building positives [11], [26], [27]. Besides, negatives should be enough to cover all information of the datasets and be consistent for adequate contrastive supervision. Momentum contrast (MoCo) [10] upkeeps a queue to collect contrastive representations and then dynamically updates the queue with current mini-batch in a first-in first-out fashion. It employs a momentum encoder to enable new negatives as consistent as possible with the existing counterparts. Simple framework for CL representation (SimCLR) [11] uses a large batch, treating all instances except a positive one in the collection as negatives, and enlarges the negative size twice with their augmented views. Although have yielded impressive results, these methods need massive computation and memory overhead or lag for updating negative adversaries and, thus, are unfriendly to challenging negative bias.

Instead of collecting contrastive samples from a memory bank or current batch, adversarial contrast (AdCo) [28] learns the negatives directly as parameters trained by maximizing contrastive loss, such as a noise-contrastive estimation-based loss (InfoNCE) [29], with advantages in updating the negatives consistently and pursuing challenging ones, for efficiency supervision. Moreover, CaCo [24] learns both the positive and negative samples directly by cooperative–adversarial training with the query encoder. Significantly, CL with learnable examples can rely less on huge batch size and strong augmentations, while excelling at tracing hard negatives. Nevertheless, their excessive uniformity of data representation on the unit hypersphere might be undesirable for keeping the intraclass manifold.

CL with single instance positives specializes in view-invariant representations. However, it neglects the relationships within a semantic class, leading to a break of intraclass diversity when catching the interclass divergence. To prevent trivial sampling of positives and preserve more variances within the intraclass, recent works proceed from diverse positive instances sampling or consistency regularization. For instance, nearest-neighbour CL representations (NNCLR) [30] replaces single instance positives with nearest neighbors, hierarchical contrastive selective coding (HCSC) [31] selects more positives when they are with similar hierarchical semantics, adaptive soft CL (ASCL) [32] returns one-hot label in instance discrimination with a soft label, and relational SSL (ReSSL) [33] adds relative consistency regularization on contrastive loss to soften relationships between different instances.

B. Challenges in Contrastive Sampling for HSIC

Many augmentation methods have been tried for effective contrastive representation learning on HSIC, such as Gaussian noise [17], random crop [20], multistrategy hybrid [21], and others. However, HSI records abundant spatial and spectral signatures for scene recognition, compared with color (RGB) images. The aforementioned methods focus much on spatial semantic learning but put aside some critical information hidden inside the spectrum. Thus, cross-domain CL in [22] treats spectral and spatial features from the target as positive pairs and performs domain-invariant representation with two independent encoders. ContrastNet [18] employs the variational AE (VAE) and adversarial AE (AAE) as augmentation functions to learn positive pairs for CL.

The high-resolution spectrum in HSI benefits fine ground recognition, such as vegetation classification. However, the data are typical in intraclass divergence and interclass confusion and sometimes with vague spatial semantics. Augmentations applicable to datasets with rich spatial semantics may be unsuitable for HSI, especially for sequential spectral data with complex manifolds that might be uncontrollable in maintaining proper mutual information. Learnable contrastive sampling facilitates compact alignment and discrimination, reducing the demand for data augmentations and huge negatives. However, it will easily result in overclustering and trivial intraclass sampling, particularly for datasets in strong nonlinearity. In this work, we will alleviate those problems by suppressing fake clusters and enriching the intracluster semantics.

III. Proposed Method

In this section, we first overview CL with learnable sampling for HSIC. Then, we introduce our modifications, sparse sampling with KL regularizer, and multiple positives learning, to improve CL in discriminative feature representation.

A. Overview of Learnable Contrastive Sampling for HSIC

To achieve self-supervised representation, CL algorithms maintain a dictionary for anchor querying, and an encoder who tries to align the anchor with a positive prototype and push

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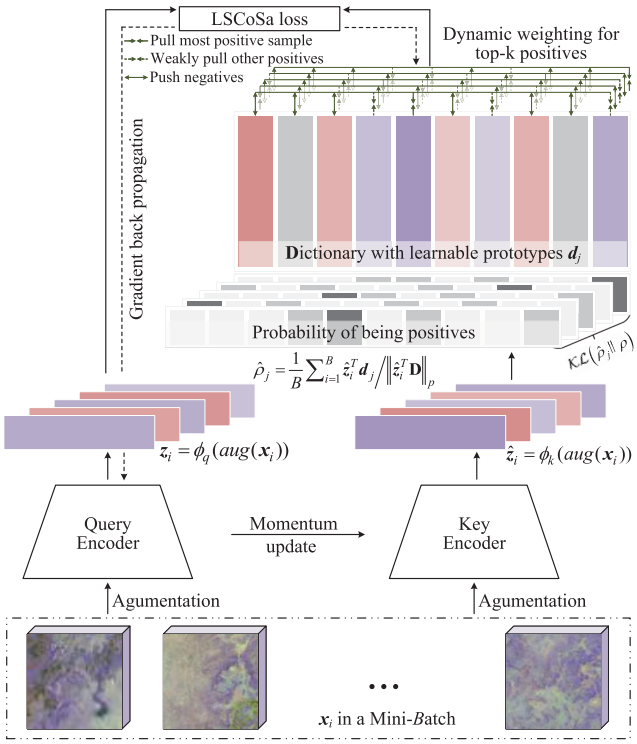


Fig. 2. Overview of our proposed LSCoSa framework. As in other contrastive SSL, LSCoSa contains two branches: a query encoder for online learning and a slowly updated key encoder offering targets. We maintain a dictionary filled with learnable prototypes for contrastive sampling. KL regularizer exerted on the being-positive probability of each prototype, measured by the average cosine similarity of key encoding with prototypes in a mini-batch, to force a few possible prototypes sampled as positive. Moreover, top-k positive prototypes are retrieved and dynamically weighted for LSCoSa loss.

it away from the other negatives in the dictionary. Similarly, our proposed LSCoSa learns representations by two encoders; one is a query encoder $\phi_q(\boldsymbol{x})$, parameterized by $\boldsymbol{\theta}_q$, and we expected for downstream task. The other key encoder $\phi_k(\boldsymbol{x})$ evolves its parameter $\boldsymbol{\theta}_k$ smoothly by exponential moving average (EMA) of $\boldsymbol{\theta}_q$, providing a dynamic dictionary for contrastive supervision, as shown in Fig. 2.

Specifically, considering a batch $\mathcal{B} = \{\boldsymbol{x}_1, \boldsymbol{x}_2, \ldots, \boldsymbol{x}_B\}$ with B unlabeled samples, we aim to learn a representation vector $\boldsymbol{z}_i \triangleq \phi_q(\text{aug}(\boldsymbol{x}_i))$ for each $\boldsymbol{x}_i \in \mathcal{B}$, where data augmentations are applied on each $\boldsymbol{x}_i$ to create a positive pair to train the encoder $\phi_q(\cdot)$. Meanwhile, the dynamic dictionary is usually built by the representation samples $\hat{\boldsymbol{z}}_i \triangleq \phi_k(\text{aug}(\boldsymbol{x}_i))$ in current batch or a queue. For convenience, we collect all the contrastive samples, positives, and negatives, in a matrix $\mathbf{D} = [\boldsymbol{d}_1, \boldsymbol{d}_2, \cdots, \boldsymbol{d}_K] \triangleq [\hat{\boldsymbol{z}}_1, \hat{\boldsymbol{z}}_2, \cdots, \hat{\boldsymbol{z}}_K]$, where $\boldsymbol{d}_{j^+}$ $(j = 1, \ldots, K)$ denotes the positive example and others denoted as $\boldsymbol{d}_j$ are negative counterparts. Then, InfoNCE loss [29] is defined to pull the positives close and push the negatives away. The encoder $\phi_q(\boldsymbol{x})$ with parameter $\boldsymbol{\theta}_q$ is trained by

$$\boldsymbol{\theta}_q^* = \arg\min_{\boldsymbol{\theta}_q} \sum_{\boldsymbol{x}_i \in \mathcal{B}} \mathcal{L}(\boldsymbol{x}_i) \tag{1}$$

where

$$\mathcal{L}(\boldsymbol{x}_i) = -log \frac{\exp\left(\boldsymbol{z}_i \cdot \boldsymbol{d}_{j^+}/\tau\right)}{\exp\left(\boldsymbol{z}_i \cdot \boldsymbol{d}_{j^+}/\tau\right) + \sum_{j \neq j^+} \exp\left(\boldsymbol{z}_i \cdot \boldsymbol{d}_j/\tau\right)}. \tag{2}$$

Dot $(\cdot)$ operator performs inner product for two normalized vectors, and τ is a temperature parameter.

CaCo [24] transfers contrastive sampling to learnable parameters, where prototypes in dictionary $\mathbf{D}$ are not collected directly from samples' representations by the key encoder, but trained along with the contrastive loss in a cooperative and adversarial fashion

$$\mathbf{D}^*, \boldsymbol{\theta}_q^* = \arg \min_{\boldsymbol{d}_{j^+}, \boldsymbol{\theta}_q} \max_{\boldsymbol{d}_{j \neq j^+}} \sum_{\boldsymbol{x}_i \in \mathcal{B}} \mathcal{L}(\boldsymbol{x}_i) \tag{3}$$

where the positive sample $\boldsymbol{d}_{j^+}$ for each $\boldsymbol{x}_i$ is chosen according to the maximum probability that the key embedding $\hat{\boldsymbol{z}}_i$ is similar with

$$\boldsymbol{d}_{j^+} = \arg\max_{\boldsymbol{d}_j} p\left(\boldsymbol{d}_j \middle| \hat{\boldsymbol{z}}_i\right) \tag{4}$$

where

$$p\left(\boldsymbol{d}_j \middle| \hat{\boldsymbol{z}}_i\right) = \frac{\exp\left(\hat{\boldsymbol{z}}_i \cdot \boldsymbol{d}_j/\tau\right)}{\sum_{j=1}^{K} \exp\left(\hat{\boldsymbol{z}}_i \cdot \boldsymbol{d}_j/\tau\right)}. \tag{5}$$

CL with learnable prototype sampling shares the dictionary over all training samples, where the prototypes can be updated on the fly. This prevents encoder optimization from being overly dependent on large batch size. Meanwhile, contrasting with prototypes rather than representations from single instance helps the model request less specific data augmentation. More importantly, adversarial training creates challenging negative prototypes, providing more discriminative information for data representation. However, the positives are chosen according to maximum similarity probability, which may be uncontrollable under unsatisfactory representation. Thus, we must initialize the dictionary carefully and set the temperature lower enough to sharpen the similarity and avoid model collapse. But, meanwhile, this causes the representations to be excessively uniform on the unit hypersphere, further confusing intraclass diversity and interclass divergence, especially undesirable for representation feature learning in HSIC. Next, we state our approach to the above limitations.

B. Sparse Sampling With KL Regularizer

Strong spectral similarity and uncertainty among objects in HSI poses challenges in effectively tracing clusters corresponding to the same category. Data representation with excessive uniformity means a truth that the fake prototypes are chosen as positives for discriminative learning, further deteriorating the intraclass manifold. Thus, prototypes retrieved with a low probability density should be suppressed sampling as positives. Here, we introduce the KL divergence regularizer on the average activation of the prototypes as positives to achieve sparse sampling.

Specifically, let $\boldsymbol{q}_i \in \mathbb{R}^K$ denote the sampling probability of each prototype $\boldsymbol{d}_j$ $(j = 1, \ldots, K)$ being positive to the target $\boldsymbol{z}_i$. The greater the similarity, the higher the probability of $\boldsymbol{d}_j$ being selected as a positive sample. Thus, we employ the cosine similarity and l_p norm to measure the chosen probability of each prototype, which can be defined as follows:

$$\boldsymbol{q}_i = \hat{\boldsymbol{z}}_i^T \mathbf{D} \Big/ \left\| \hat{\boldsymbol{z}}_i^T \mathbf{D} \right\|_p. \tag{6}$$

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The l_p norm could simulate the probability of the same sampling situation happening p times here. A larger p value could provide a more stable probability evaluation on a mini-patch but quickly fall into overconfidence. In our later experiments, we select the l_2 norm and will analyze the impact of p value on classification accuracy.

To suppress fake prototypes being activated, we try to constrain the prototypes' average chosen (or activation) probability, forcing it close to a small value. Accordingly, we define the average chosen probability of $\boldsymbol{d}_j$ on a mini-batch as follows:

$$\hat{\rho}_j = \frac{1}{B}\sum_{i=1}^{B} \boldsymbol{q}_{ij} = \frac{1}{B}\sum_{i=1}^{B} \hat{\boldsymbol{z}}_i^T \boldsymbol{d}_j \Big/ \left\| \hat{\boldsymbol{z}}_i^T \mathbf{D} \right\|_p. \tag{7}$$

Then, we penalize the Bernoulli distribution with activation probability $\hat{\rho}_j$ close to a given Bernoulli random variable with mean ρ. The KL divergence is often used to measure how a distribution matches another, where the forward KL prefers to cover all the modes by mean distribution; the reverse KL emphasizes mimicking one of the modes with high probability and wide support and prevents the approximate distribution from collapsing by entropy maximum condition. We aim to suppress the chosen probability of the prototype. Thus, the reverse KL divergence is employed here for sparsity constraint

$$\mathcal{KL}\left(\hat{\rho}_j \| \rho\right) = \hat{\rho}_j \log \frac{\hat{\rho}_j}{\rho} + (1-\hat{\rho}_j)\log \frac{1-\hat{\rho}_j}{1-\rho} \tag{8}$$

where ρ is a given sparsity parameter, to control the average activation. Our subsequent experiments further illustrate that reverse KL divergence has a more friendly sparsity than the forward process.

Minimizing the sparsity penalty term can prevent all the prototypes $\boldsymbol{d}_j$ from being activated frequently. Thus, the loss function of contrastive SSL with sparse sampling is now

$$\begin{aligned} &\arg\min_{\boldsymbol{\theta}_q, \boldsymbol{d}_j} \sum_{\boldsymbol{x}_i \in \mathcal{B}} \mathcal{L}(\boldsymbol{x}_i) + \lambda \sum_{j=1}^{K} \mathcal{KL}\left(\hat{\rho}_j \| \rho\right) \\ &\text{s.t. } \mathbf{D}^* = \sum_{\boldsymbol{x}_i \in \mathcal{B}} \min_{\boldsymbol{d}_{j^+}} \max_{\boldsymbol{d}_{j \neq j^+}} \mathcal{L}(\boldsymbol{x}_i) \end{aligned} \tag{9}$$

where λ controls the weight of the KL penalty term. We optimize the parameters $\boldsymbol{\theta}_q$ of the query encoder and prototypes $\boldsymbol{d}_j$ in the dynamic dictionary through gradient descent by backpropagation of the loss function, and they are updated alternately with cooperative–adversarial learning of positive and negative samples as CaCo does in [24].

As illustrated in Fig. 1(b), the KL penalty significantly suppresses the activation of prototypes, and only a portion of them are selected as positives. This finding demonstrates a significant advantage in linear evaluation. HSIs have extreme spectral variation within the same category. Meanwhile, the alignment with single instance positives may easily fall into trivial clusters, although sparse sampling improved this phenomenon indirectly. Thus, we further propose multiple positives assistant learning in keeping semantic diversity.

C. Multiple Positives Learning

CL with InfoNCE loss [29] is instance-level discrimination, where one-hot pseudo-label $\boldsymbol{y} \in \mathbb{R}^K$ is generated by key $\hat{\boldsymbol{z}}_i$ matching with the most similar prototype $\boldsymbol{d}_j$, and anchor $\boldsymbol{z}_i$ querying in the dynamic dictionary measures prediction probability. Then, cross-entropy loss with a K-way softmax-based classifier is employed to classify $\boldsymbol{z}_i$ into the cluster with prototype $\boldsymbol{d}_j$. Thus, we rewrite (2) as follows:

$$\mathcal{L}(\boldsymbol{x}_i) = -\sum_{j=1}^{K} \boldsymbol{y}_j \log p\left(\boldsymbol{d}_j | \boldsymbol{z}_i\right) \tag{10}$$

where

$$\boldsymbol{y}_j = \begin{cases} 1, & j = \arg\max_j p\left(\boldsymbol{d}_j | \hat{\boldsymbol{z}}_i\right) \\ 0, & \text{otherwise.} \end{cases} \tag{11}$$

As mentioned above, the one-hot contrastive loss is unfriendly to the intraclass spectral variant and encourages trivial clustering because of overconfidence in discriminating negative examples. Thus, we propose multiple positives learning, where the top-k most potential prototypes are searched as positives for CL. We collect those positive examples into $\mathcal{N}_k^+$; then, (11) can be rewritten as follows:

$$\boldsymbol{y}_j = \begin{cases} 1/k, & j \in \mathcal{N}_k^+ \\ 0, & \text{otherwise.} \end{cases} \tag{12}$$

Equation (12) treats the top-k positive examples equally with the same hard label. This will sacrifice the anchor alignment when minimizing cross-entropy loss. Besides, it, in turn, causes contrastive sampling in low probability and density estimation in high uncertainty, because we share the same function in estimating pseudo-label and predicting categorical probability. To accomplish anchor alignment and confident clustering in rich intracluster semantics simultaneously, we treat the top-k positives differently, assigning the most one with weight 1 for anchor alignment and dynamic weighting the others by a much smaller value α to weakly minimize the distance, keeping nontrivial clusters. Thus, we redefine the cross-entropy loss as follows:

$$\mathcal{L}_M(\boldsymbol{x}_i) = -\sum_{j=1}^{K} \frac{\boldsymbol{y}_j}{\|\boldsymbol{y}\|_1} \log p\left(\boldsymbol{d}_j | \boldsymbol{z}_i\right) \tag{13}$$

where $\boldsymbol{y}_j$ is normalized by l_1 norm and defined as follows:

$$\boldsymbol{y}_j = \begin{cases} 1, & j = \arg\max_j p\left(\boldsymbol{d}_j | \hat{\boldsymbol{z}}_i\right) \\ \alpha, & j \neq \arg\max_j p\left(\boldsymbol{d}_j | \hat{\boldsymbol{z}}_i\right) \text{ and } j \in \mathcal{N}_k^+ \\ 0, & j \notin \mathcal{N}_k^+. \end{cases} \tag{14}$$

Intracluster diversity always confuses with pursuing hard negatives. Consequently, when all the prototypes are highly uncertain about being sampled as positives, we should set the $k-1$ positives with a smaller α, to avoid treating hard negatives as positives arbitrarily. In light of this, we expect the α to vary inversely as the entropy on the average chosen probability

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Algorithm 1 LSCoSa

Inputs:
query encoder ϕ_q and initialized parameter $\boldsymbol{\theta}_q$
key encoder ϕ_k and initialized parameter $\boldsymbol{\theta}_k$
initialized dictionary $\mathbf{D}$ of size K
Batch size N, momentum m, temperature τ, sparsity ρ
for sampled mini batch $\mathcal{B}$ **do**
for all $\boldsymbol{x}_i$ in batch $\mathcal{B}$ **do**
$\boldsymbol{z}_i = \phi_q(aug(\boldsymbol{x}_i))$
$\hat{\boldsymbol{z}}_i = \phi_k(aug(\boldsymbol{x}_i))$
$\boldsymbol{q}_i = \hat{\boldsymbol{z}}_i^T \mathbf{D} \Big/ \left\| \hat{\boldsymbol{z}}_i^T \mathbf{D} \right\|_p$
$\hat{\rho}_j = \hat{\rho}_j + \boldsymbol{q}_{ij}, \quad j = 1, 2, \ldots K$
$\mathcal{N}_{k^+} \leftarrow$ index j of top-k prototypes
end for
$\hat{\rho}_j = \hat{\rho}_j / B, \quad j = 1, 2, \ldots K$
$\mathcal{KL}(\hat{\rho}_j \| \rho) = \hat{\rho}_j \log \frac{\hat{\rho}_j}{\rho} + (1 - \hat{\rho}_j) \log \frac{1-\hat{\rho}_j}{1-\rho}$
$\alpha = \frac{1}{1+e^{H(\hat{\rho})}}$, where $H(\hat{\rho}) = -\sum_{j=1}^{K} \hat{\rho}_j \log(\hat{\rho}_j)$
$\mathcal{L}_M(\boldsymbol{x}_i) = -\sum_{j=1}^{K} \frac{\boldsymbol{y}_j}{\|\boldsymbol{y}\|_1} \log p(\boldsymbol{d}_j | z_i)$, $\boldsymbol{y}_j$ is as in Eq. 14
$\mathcal{L} = \sum_{\boldsymbol{x}_i \in \mathcal{B}} \mathcal{L}_M(\boldsymbol{x}_i) + \lambda \sum_{j=1}^{K} \mathcal{KL}(\hat{\rho}_j \| \rho)$
update ϕ_q and $\mathbf{D}$ to minimize $\mathcal{L}$ and cooperative-adversarial of the subjection in Eq. 17
$\boldsymbol{\theta}_k = m\boldsymbol{\theta}_k + (1-m)\boldsymbol{\theta}_q$
end for
Output: encoder ϕ_q

of all the prototypes. So, we define the dynamic weight as follows:

$$\alpha = \frac{1}{1 + e^{H(\hat{\rho})}} \tag{15}$$

where $\hat{\boldsymbol{\rho}} = [\hat{\rho}_1, \hat{\rho}_2, \cdots, \hat{\rho}_K]$, and

$$H(\hat{\rho}) = -\sum_{j=1}^{K} \hat{\rho}_j \log(\hat{\rho}_j). \tag{16}$$

The greater the entropy $H(\hat{\rho})$, the more uncertain the prototypes sampling as positives; thus, a smaller weight α being set to those assistant positives. By now, the loss function of LSCoSa can be rewritten as follows:

$$\begin{aligned} &\arg\min_{\boldsymbol{\theta}_q, \boldsymbol{d}_j} \sum_{\boldsymbol{x}_i \in \mathcal{B}} \mathcal{L}_M(\boldsymbol{x}_i) + \lambda \sum_{j=1}^{K} \mathcal{KL}(\hat{\rho}_j \| \rho) \\ &\text{s.t. } \mathbf{D}^* = \sum_{\boldsymbol{x}_i \in \mathcal{B}} \min_{\boldsymbol{d}_{j^+}} \max_{\boldsymbol{d}_{j \neq j^+}} \mathcal{L}(\boldsymbol{x}_i). \end{aligned} \tag{17}$$

It should be noted that we keep updating the prototypes $\boldsymbol{d}_{j \neq j^+}$ by adversarial training as the negative examples do, even when they are sampled as the $k-1$ assistant positives. This means we select the overhard negatives as latent positives to disturb the most aligned prototype to a good generalization and to keep all the clusters in semantic diversity. In addition, we update the loss function as mentioned previously for (13). Algorithm 1 summarizes our proposed method LSCoSa.

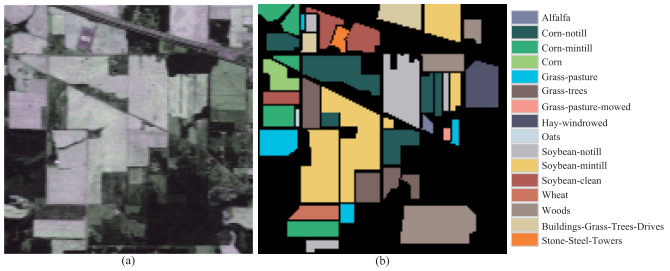


Fig. 3. IP dataset. (a) False-color image. (b) Ground truth.

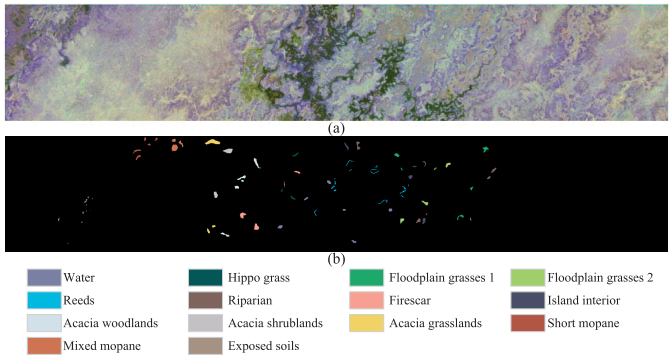


Fig. 4. BO dataset. (a) False-color image. (b) Ground truth.

IV. Experiments and Analysis

In this section, we conduct a comprehensive set of experiments to evaluate our LSCoSa, including ablation studies of the major components, parameter sensitivity analysis, and comparison with some state-of-the-art classification models.

A. Experimental Datasets

We perform experiments on three benchmark datasets, including IP, Botswana (BO), and HyRANK-Dioni (HR-D).

1) Indian Pines: The IP dataset, collected by the Airborne Visible Infrared Imaging Spectrometer (AVIRIS) sensor over northwest Indiana, consists of 145×145 pixels and 200 spectral bands. The available ground truth contains 16 land-cover classes. Fig. 3(a) and (b) shows its false-color image and the ground-truth map, respectively.

2) Botswana: The BO dataset, acquired by the Earth Observing (EO-1) satellite of National Aeronautics and Space Administration (NASA) over the Okavango Delta in BO, consists of 1476×256 pixels and 145 spectral bands. The available ground truth contains 14 classes. Fig. 4(a) and (b) shows its false-color image and the ground-truth map, respectively.

3) HyRANK-Dioni: The HR-D dataset, one of the five satellite HSIs from the EO-1 Hyperion sensor, consists of 250×1376 pixels and 176 spectral bands. The available ground truth contains 12 classes. Fig. 5(a) and (b) shows its false-color image and the ground-truth map, respectively.

B. Implementation Details

We adopt a ResNet [34] backbone as the query encoder for data representation, which contains one convolutional layer of $7 \times 7 \times 64$ kernel size and four building blocks with the

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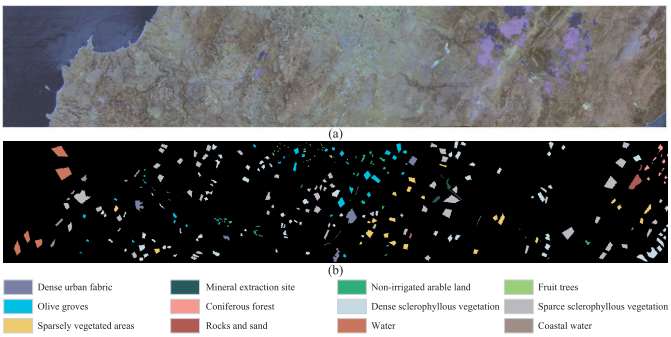


Fig. 5. HR-D dataset. (a) False-color image. (b) Ground truth.

TABLE I
LINEAR EVALUATION RESULTS OF LSCOSA TRAINED WITH FORWARD VERSUS REVERSE KL DIVERGENCE

Forward	Reverse	IP	BO	HR-D
	✓	**96.70**	**99.04**	**95.29**
✓		95.09	98.24	94.53
CaCo [24]		92.68	96.91	93.95

same filter size $3 \times 3 \times 128$. Then, the 128-D embedding by global average pooling is used for CL. We share this backbone with the key encoder but update its parameters with a large momentum ($m = 0.99$ in experiments). All samples from one dataset are used for model pretraining.

We pretrain the encoder for 200 epochs with a batch size of 32. Layerwise adaptive rate scaling (LARS) [35] is used as an optimizer to update the parameters with a momentum of 0.99 and a weight decay of 1.5×10^{-6}. We update the dictionary with the stochastic gradient descent (SGD) [36] optimizer with a learning rate of 0.02 and a momentum of 0.9. We set the temperature $\tau = 0.08$ for dictionary update and InfoNCE loss.

We evaluate the learned features by a linear classifier with only one initial fully connected layer. A random selection of 10%, 3%, and 3% of samples from the IP, BO, and HR-D datasets, respectively, is chosen per class to train the classifiers, while the remaining samples are used for testing. We use the indicators, such as overall accuracy (OA), average accuracy (AA), and kappa statistic (κ), to evaluate the classification performance and provide OA results for parameter analysis and ablation studies. All our results are reported as the average of ten runs. All experiments are implemented on a PyTorch platform using a workstation with an i9-10900K CPU and an NVIDIA GeForce RTX 3090 graphics card.

C. Ablation Study

In LSCoSa, two core contributions are designed for discriminative feature learning, including KL regularizer for sparse positive sampling and multiple positives to keep representation with semantic diversity. This section discusses the critical factors in both components through ablation studies.

1) Effect of KL Divergence Types: KL regularizer imposes restrictions on the prototypes to achieve sparse positive sampling. There are two types of KL divergence: forward KL divergence and reverse KL divergence. We compare their linear probing results in Table I. They all boost LSCoSa and achieve higher accuracy over CaCo [24]. Reverse KL divergence generates lower and more friendly sparsity for density estimation, thus reaching about 1% more effective results than the forward types. Therefore, we adopt the reverse KL divergence for the subsequent experiments.

TABLE II
LINEAR EVALUATION FOR KL REGULARIZER WITH DIFFERENT NORMALIZATION FUNCTIONS

Datasets	Softmax	l_p-norm		
		$p = 1$	$p = 2$	$p = 3$
IP	93.11	95.59	**96.70**	95.71
BO	93.72	97.64	**99.04**	98.83
HR-D	93.62	94.07	95.29	**95.67**

TABLE III
LINEAR EVALUATION FOR POSITIVES WEIGHTED WITH HARD LABEL VERSUS DYNAMIC WEIGHTS

Hard	Dynamic	IP	BO	HR-D
✓		92.72	97.24	91.48
	✓	**96.70**	**99.04**	**95.29**
CaCo [24]		92.68	96.91	93.95

2) Effect of Normalization Function: KL regularizer in LSCoSa relies heavily on a well-defined normalization function to evaluate what possible probability each prototype will be sampled as positives [as in (6)]. Here, we investigate how different normalization functions, such as softmax and l_p norm with $p = 1, 2$, or 3, affect the encoder performance. Table II shows the OA results. l_p norm produces sharper sampling distribution probability, thus being 2% better in linear classification than softmax function. A larger p value achieves a higher testing OA than the smaller ones, but it is inadvisable to set an overlarge value. We chose the l_2 norm in our later experiments.

3) Effect of Weighting for Positives: We conduct an ablation study on LSCoSa to compare the contribution of weighting the multiple positives by the hard label [as (12)] versus dynamic weights [as (14)]. From the linear evaluation results in Table III, we can find that the hard label presents only slightly better results than CaCo [24] and even obtains worse results by 2% points on the HR-D dataset. The dynamic weights, in contrast, achieve significant improvement over CaCo [24] by more than 2%, particularly over by 4% on the IP dataset. This confirms that it is crucial to distinguish the assistant positives from the most similar ones and set them with small weights, benefitting diversity within the cluster.

4) Effect of Both Contributions: Finally, we conduct the ablation to our two main contributions, KL regularizer and sampling of multiple positives, and investigate how they improve LSCoSa over CaCo [24]. Table IV shows the ablation experiments; CaCo [24] corresponds to the situation where no one component is present, and LSCoSa is on the contrary. The results show that both contributions can improve the classification accuracy of the IP and BO datasets alone, and their joint presence generates superior performance. Although

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TABLE IV
ABLATION STUDIES FOR TWO COMPONENTS IN LSCOSA UNDER LINEAR PROBING ACCURACY

KL regularizer	Multiple positives	IP	BO	HR-D
		92.68	96.91	93.95
✓		94.77	98.75	93.93
	✓	94.95	96.95	93.95
✓	✓	**96.70**	**99.04**	**95.29**

TABLE V
EFFECT OF PROTOTYPE NUMBER K FOR LSCOSA UNDER LINEAR PROBING ACCURACY

Num. of prototypes	$K = 128$	$K = 256$	$K = 512$
IP	94.73	96.70	**96.91**
BO	98.26	99.04	**99.28**
HR-D	94.97	95.29	**95.82**

TABLE VI
EFFECT OF SPARSITY PARAMETER ρ IN KL DIVERGENCE FOR LSCOSA UNDER THE LINEAR PROBING ACCURACY

Sparsity	$\rho = 0.04$	$\rho = 0.2$	$\rho = 0.4$	$\rho = 0.6$	$\rho = 0.8$
IP	95.92	95.26	95.87	**96.70**	96.19
BO	97.25	97.08	**99.04**	97.18	96.54
HR-D	94.50	95.05	**95.29**	94.34	94.72

TABLE VII
EFFECT OF TRADE-OFF PARAMETER λ ON KL REGULARIZER FOR LSCOSA UNDER THE LINEAR PROBING ACCURACY

Trade-off	$\lambda = 0.4$	$\lambda = 0.8$	$\lambda = 1.2$	$\lambda = 1.6$	$\lambda = 2$
IP	95.50	95.27	96.21	**96.70**	94.98
BO	96.96	97.72	98.33	**99.04**	97.69
HR-D	**95.29**	94.74	94.53	94.94	94.00

TABLE VIII
EFFECT OF TEMPERATURE τ IN LOSS FUNCTION UNDER LINEAR PROBING ACCURACY

Temperature	$\tau = 0.01$	$\tau = 0.05$	$\tau = 0.08$	$\tau = 0.3$	$\tau = 0.5$	$\tau = 0.8$
IP	93.44	**97.29**	96.70	89.29	88.80	88.20
BO	95.93	96.65	**99.04**	97.10	96.34	93.45
HR-D	90.30	**95.31**	95.29	86.26	86.73	80.77

TABLE IX
EFFECT OF THE NUMBER k OF POSITIVE EXAMPLES FOR LSCOSA UNDER THE LINEAR PROBING ACCURACY

Num. of Positives	$k = 1$	$k = 2$	$k = 3$	$k = 4$	$k = 8$
IP	94.77	95.59	95.48	**96.70**	94.97
BO	98.75	98.64	98.61	**99.04**	96.22
HR-D	93.93	94.72	95.08	**95.29**	94.84

any single component shows either mediocre or sometimes even slightly worse results on the HR-D dataset, we can benefit remarkably from their cooperation.

D. Parameter Analysis

1) Number of Prototypes K in the Dictionary: We investigate how the performance changes, as we vary the number of prototypes K in the dictionary from 128 to 512. As shown in Table V, our approach benefits from larger dictionary sizes over smaller ones, and the OA results increase significantly from 128 to 256 and then tend to be stable. This verifies that a larger dictionary provides richer information for data representation, and the accuracy will not be affected even if it is saturated. We set $K = 256$ in other experiments, considering additional compute overhead while training.

2) Sparsity Parameter ρ in KL Divergence: We test the impact of the sparsity parameter ρ in the KL regularizer when it ranges from 0 to 1 by a stride of 0.2, except for replacing 0 with 0.04; the larger the value, the weaker the sparsity sampling. As shown in Table VI, the classification accuracy is sensitive to this parameter. But, it seems to follow the rule that datasets with more complex separability prefer a larger sparsity value, because we cannot simply suppress prototypes when the low-density region is very subtle to trace. Still, it is not the larger, the better, illustrating that the sparsity sampling is indispensable in LSCoSa.

3) Trade-Off Parameter λ in Loss Function: We next study the impact of the trade-off parameter λ between InfoNCE loss and KL regularizer in the final loss function of LSCoSa, where the λ value ranges from 0.4 to 2 by a stride of 0.4; the larger the value, the higher the proportion to KL regularizer. From the results in Table VII, it could be observed that LSCoSa performs well on IP and BO datasets when setting a high-proportion KL penalty, whereas a low-proportion KL restraint is friendly to HR-D. Likewise, too large a λ will disturb the discriminative supervision of contrastive loss.

4) Temperature τ in Loss Function: We further test the importance of temperature τ in InfoNCE loss when it ranges from 0.01 to 0.8. The lower the temperature, the greater the penalty imposed on hard negatives by InfoNCE loss. As shown in Table VIII, LSCoSa prefers relatively low temperatures, around 0.08, and the linear probing results decrease markedly when larger than 0.3. This illustrates that, as all of the contrastive samples are learned along with encoder training, a low temperature is well situated in InfoNCE loss to push away the hard negatives, ensuring stable positive sampling and preventing the encoder from collapsing.

5) Number of Positives k in $\mathcal{L}_M(\boldsymbol{x}_i)$: To study how the encoder performance changes with the number of positive samples, we compare the linear probing accuracy of LSCoSa when the number of positives ranges from 1 to 8. As in Table IX, $k = 1$ means that only the most similar prototype is retrieved as the positive sample for CL or means CaCo [24] modified with KL regularizer, and the other situations correspond to (14). The results show that relatively more positives ($k = 4$, for example) could benefit LSCoSa in maintaining semantic diversity for feature learning and, thus, better classification accuracy. Of course, there cannot be too many ($k = 8$, for instance) assistant positives, which may reduce the performance when some potential hard negative samples are treated as positives.

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TABLE X
LINEAR PROBING RESULTS WITH 10% TRAINING SAMPLES ON THE IP DATASET. THE BEST ACCURACIES ARE MARKED IN BOLD

Class	PCA	3D-CAAE	SSCL	ContrastNet	BYOL-T	S3FN	SC-SS-MTr	CaCo	LSCoSa
1	63.03	55.12	54.66	98.78	92.73	92.20	**100**	92.11	96.72
2	78.18	90.80	90.11	91.41	92.19	88.15	94.49	92.33	**96.76**
3	77.83	88.33	75.52	88.54	86.32	93.31	94.14	88.01	**95.11**
4	60.84	76.43	84.82	85.49	92.98	94.74	**99.94**	98.36	95.72
5	88.39	90.16	88.97	96.18	96.60	91.70	96.75	95.77	**99.12**
6	93.52	95.33	85.34	95.18	91.06	97.69	**99.32**	93.38	97.90
7	56.95	62.80	93.62	**100**	96.44	89.20	**100**	90.91	**100**
8	92.80	95.40	98.89	98.40	99.88	99.09	**100**	99.42	**100**
9	15.00	18.33	34.25	86.11	39.66	42.22	**100**	37.84	56.97
10	75.32	92.25	86.48	92.72	87.35	93.21	**94.52**	83.06	91.17
11	86.56	95.54	93.38	94.24	96.16	95.73	91.55	95.37	**98.06**
12	69.55	81.09	84.71	91.31	89.04	93.00	94.06	92.21	**95.92**
13	93.44	92.81	92.59	90.49	98.03	97.89	**99.86**	99.46	96.94
14	93.67	97.41	95.13	98.63	98.41	98.28	**98.94**	96.25	98.77
15	71.20	86.89	58.25	94.15	76.15	94.21	**99.81**	78.57	91.57
16	85.19	67.26	88.50	85.48	85.40	98.45	**99.66**	93.75	93.89
OA(%)	83.06	91.61	88.53	93.62	92.80	94.35	95.36	92.68	**96.70**
AA(%)	75.09	80.37	81.58	92.94	88.65	91.19	**97.69**	89.18	94.04
Kappa(%)	80.77	90.45	86.87	92.72	91.77	93.56	94.67	91.64	**96.23**

TABLE XI
LINEAR PROBING RESULTS WITH 3% TRAINING SAMPLES ON THE BO DATASET. THE BEST ACCURACIES ARE MARKED IN BOLD

Class	PCA	3D-CAAE	SSCL	ContrastNet	BYOL-T	S3FN	SC-SS-MTr	CaCo	LSCoSa
1	94.26	91.79	92.29	87.02	95.58	**100**	85.17	96.68	97.54
2	78.05	90.20	62.62	87.76	98.62	**100**	95.74	93.01	99.49
3	88.20	94.98	97.93	94.65	98.87	97.94	86.72	**100**	**100**
4	87.69	93.41	86.08	98.56	97.63	98.56	99.09	98.10	**100**
5	78.98	80.69	81.04	84.29	88.21	89.66	86.87	93.10	**94.69**
6	81.53	85.48	82.66	98.85	99.19	80.84	85.92	90.65	**99.54**
7	94.30	91.55	95.47	99.00	**100**	99.60	99.92	**100**	**100**
8	94.44	85.94	87.78	98.98	99.95	**100**	91.99	99.98	**100**
9	95.72	94.92	97.18	**100**	97.61	99.02	**100**	99.01	**100**
10	55.54	95.23	89.76	**100**	97.25	**100**	**100**	**100**	**100**
11	89.19	96.15	90.36	95.95	99.38	97.64	95.20	99.00	**100**
12	73.18	92.33	87.30	98.86	94.47	94.89	**100**	88.75	**100**
13	87.97	90.12	83.37	**100**	99.31	99.62	**100**	99.31	**100**
14	85.11	82.07	89.01	80.43	**90.48**	82.61	90.23	**90.48**	**90.48**
OA(%)	86.55	90.71	88.75	95.49	97.18	96.13	94.05	96.91	**99.04**
AA(%)	84.58	90.15	87.35	94.61	96.90	95.74	94.06	96.29	**98.70**
Kappa(%)	85.33	89.94	87.80	95.12	96.94	95.80	93.56	96.65	**98.96**

E. Comparison With Other Methods

We compare our proposed model LSCoSa with some feature learning methods for HSIC, such as a semisupervised 3-D convolutional adversarial AE model (3D-CAAE) [37], two SimCLR-based models that one uses spatial augmentation as positive pair (SSCL) [17] and the other learns two different representations as positives and combines the InfoNCE and prototype contrastive loss functions (ContrastNet) [18], BYOL-T [19], a self-supervised spectral–spatial feature learning network (S3FN) with contrastive representation alignment [38], a hybrid model comprising label supervision, mask reconstruction, and representation CL (SC-SS-MTr) [39], and CaCo [24] that have the same backbone as LSCoSa for reference. Besides, we report the HSIC results by radial basis function-based support vector machine (RBF-SVM) with the first 15 components of principal component analysis (PCA) as data representation, a baseline for performance evaluation.

Tables X–XII report the mean results of ten runs from all the methods on the IP, BO, and HR-D datasets, including the accuracy per class and the overall quantifications. We set the same input patch size of 15×15 and employ the same linear evaluation protocol for all the SSL frameworks for fairness. The other parameters involved are set according to the corresponding references.

From the base result of PCA, we can confirm that all the self-supervised pretraining models improve the data representation, except SSCL does slightly worse on the HR-D dataset. The 3D-CAAE [37] does not achieve competitive results, even though it employs a small number of manual label information. The results from two SimCLR-based models illustrate that spatial augmentation (as in SSCL [17]) is inappropriate for HSI datasets with vague spatial semantics. ContrastNet [18] pretrains the positives by two generative encoder networks (VAE and AAE), achieving significant and stable

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TABLE XII
LINEAR PROBING RESULTS WITH 3% TRAINING SAMPLES ON THE HR-D DATASET. THE BEST ACCURACIES ARE MARKED IN BOLD

Class	PCA	3D-CAAE	SSCL	ContrastNet	BYOL-T	S3FN	SC-SS-MTr	CaCo	LSCoSa
1	74.33	83.66	75.75	83.43	80.41	73.28	**88.37**	86.56	87.25
2	68.45	72.73	84.09	70.86	96.62	98.99	**100**	98.41	97.27
3	69.12	70.37	54.60	53.74	74.66	**89.09**	82.27	80.46	85.97
4	14.40	48.90	58.16	33.77	52.36	70.55	**92.00**	82.16	81.03
5	84.69	86.59	73.99	86.15	82.76	84.14	90.51	**91.64**	91.59
6	88.29	95.60	94.25	91.03	97.18	66.29	**100**	99.04	**100**
7	89.74	91.74	90.00	89.00	91.32	91.58	92.14	95.66	**96.19**
8	88.33	91.41	87.84	89.18	89.38	95.96	90.99	95.45	**96.51**
9	77.85	91.79	78.88	90.45	80.20	93.18	**96.60**	89.54	95.85
10	83.33	93.58	85.42	94.13	92.47	93.50	**99.10**	90.58	93.74
11	99.47	**100**	96.57	99.81	**100**	**100**	99.74	99.83	**100**
12	94.60	98.81	82.00	97.51	99.87	94.82	**100**	99.28	99.35
OA(%)	86.10	89.52	85.03	88.21	88.62	91.48	92.59	93.95	**95.29**
AA(%)	77.72	85.31	80.13	81.63	86.44	87.62	**94.31**	92.38	93.69
Kappa(%)	82.93	87.08	81.42	85.42	85.71	89.41	90.82	92.49	**94.14**

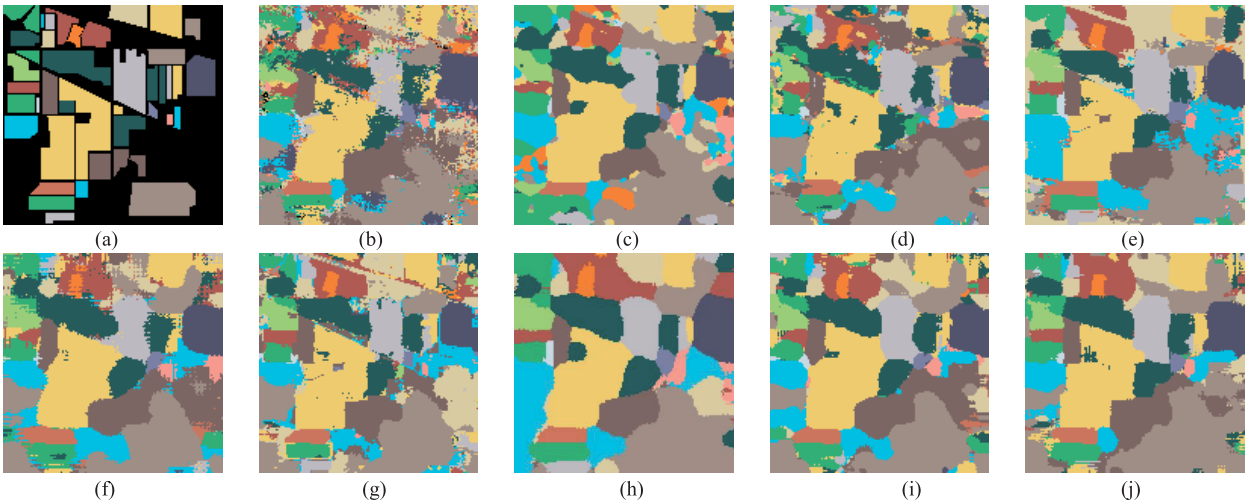


Fig. 6. Classification maps of the IP dataset produced by (a) ground truth, (b) PCA, (c) 3D-CAAE [37], (d) SSCL [17], (e) ContrastNet [18], (f) BYOL-T [19], (g) S3FN [38], (h) SC-SS-MTr [39], (i) CaCo [24], and (j) LSCoSa.

improvements on all the testing datasets. However, it benefits only the IP dataset when compared with BYOL-T. BYOL [40] focuses on uniformity of data representation and introduces EMA, predictor, and batch normalization to prevent the contrastive model from collapsing. Since independent of negative samples, it requires less careful data augmentation. Thus, BYOL-T [19] performs much better than the SimCLR-based methods. With the same emphasis on representation uniformity, CaCo [24] presents almost the same results as BYOL-T [19] and is outstanding on the HR-D dataset where hard negative supervision underlines significant advantages. Benefiting from contrastive feature alignments within specific spectral and spatial features and between them, S3FN [38] shows more comparable performance than BYOL-T [19] on IP and HR-D datasets. However, it cannot maintain a stable advantage, such as the results from the BO dataset. A similar situation happened in the results from SC-SS-MTr [39], even though label supervision and mask reconstruction were intervened.

As previously stated, excessive representation uniformity might be unfriendly to data manifold. LSCoSa improves CaCo [24] by sparse sampling and multiple positives assistant, performing as 4.02%, 2.13%, and 1.34% OA improvement on the testing datasets, respectively. LSCoSa also achieves the best results in most categories on the three datasets, especially in BO. The IP dataset is typical in class imbalance and spectral uncertainty. The ninth class, for instance, takes a much smaller land surface and has only 20 labeled samples, which is easily overlooked in batch supervision. LSCoSa considers the imbalance class well, although it does not exceed ContrastNet [18] and SC-SS-MTr [39], which perform data representation with additional generative models. Because of label supervision and mask reconstruction intervention, SC-SS-MTr [39] performs outstandingly in classes with extremely few samples, thus achieving the best AA in IP and HR-D datasets, but lower OA and kappa statistics. All these results assure that our model performs excellently in generalization representation with self-supervised feature learning.

Furthermore, we show all the classification maps in Figs. 6–8. An ideal data representation can achieve HSIC with intraclass smoothness and accurate boundary positioning. From the results, we can find that our LSCoSa exhibits more

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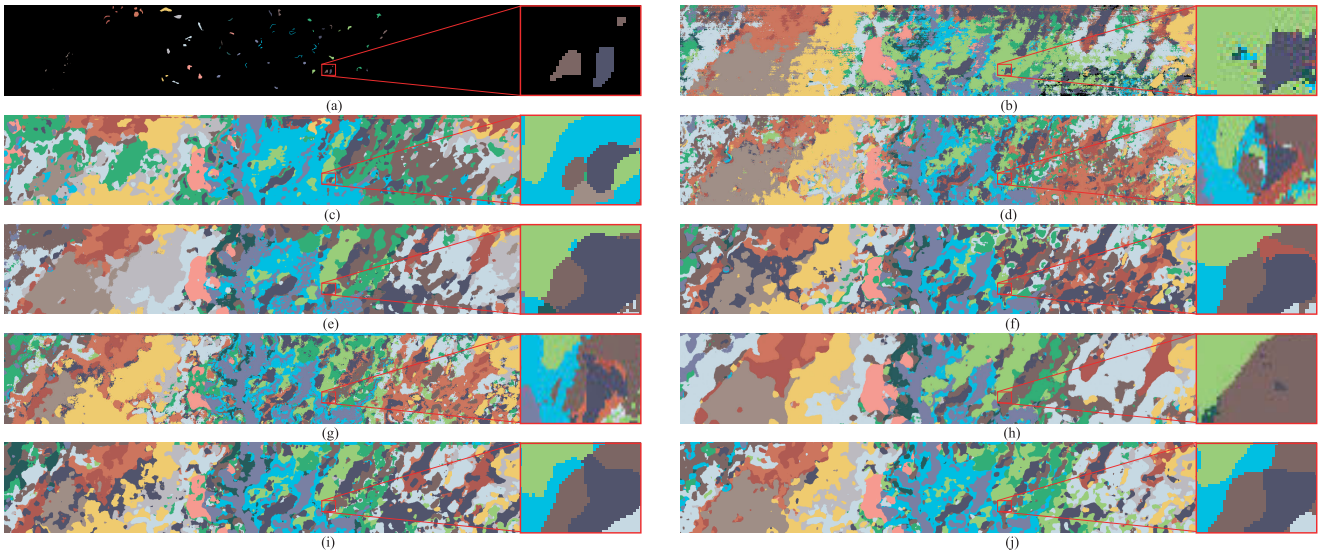


Fig. 7. Classification maps of the BO dataset produced by (a) ground truth, (b) PCA, (c) 3D-CAAE [37], (d) SSCL [17], (e) ContrastNet [18], (f) BYOL-T [19], (g) S3FN [38], (h) SC-SS-MTr [39], (i) CaCo [24], and (j) LSCoSa.

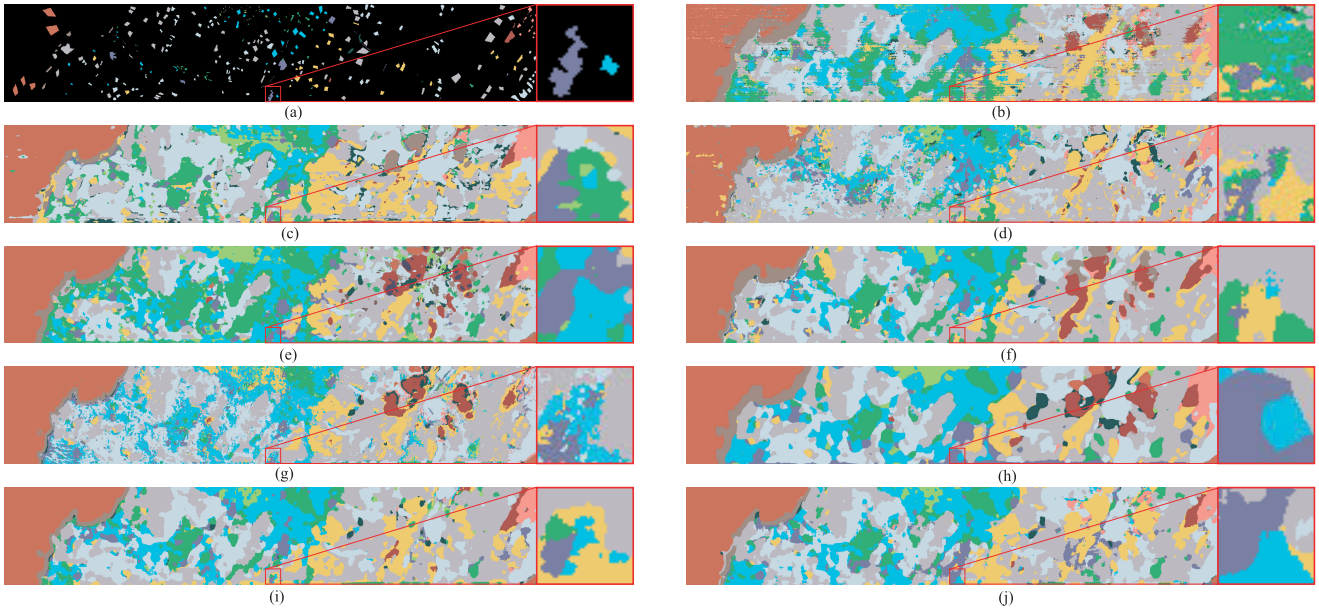


Fig. 8. Classification maps of the HR-D dataset produced by (a) ground truth, (b) PCA, (c) 3D-CAAE [37], (d) SSCL [17], (e) ContrastNet [18], (f) BYOL-T [19], (g) S3FN [38], (h) SC-SS-MTr [39], (i) CaCo [24], and (j) LSCoSa.

consistent regional categories and sharper boundaries as a whole. As for the IP dataset, although LSCoSa misclassifies some small area targets, such as category 9 (Oats with only 20 samples), it offers more precise boundary positioning compared with SC-SS-MTr [39]. More importantly, for the BO and HR-D datasets with more granular targets, all comparison methods are either sensitive to noise and cause serious misclassification, or overly smooth in distinguishing neighborhoods and overwhelm classification boundaries. Remarkably, LSCoSa balances these two aspects, achieving better intraclass smoothness and more accurate boundary location.

SSL aims to learn high-level semantic information from the wealth of unlabeled data and to alleviate overfitting caused by a few training samples. Accordingly, we compare the feature learning performance of our proposed LSCoSa with all the other methods by linear probing when the training samples per class range from 1% to 15%. From the results shown in Fig. 9, we can find that our LSCoSa is less than satisfactory on IP and BO datasets under 1% training samples. That is probably because those two datasets contain targets with highly similar spectra, and instance-level CL will easily cause overuniformity of representation distribution, creating obstacles to linear probing. Coincidentally, except for SC-SS-MTr [39] benefited from label supervision and mask reconstruction, all these methods with higher accuracy, such as ContrastNet [18], BYOL-T [19], and S3FN [38], start with PCA preprocessing, which seems to be helpful for feature learning, but it limits the model's upper bound of the optimal performance. Nevertheless, LSCoSa strives to alleviate the overuniformity phenomenon and for a diversity of semantic representation, achieving an increase of over 2% compared with CaCo [24]. It is further worth noting that once a few more samples, more than 2% per class,

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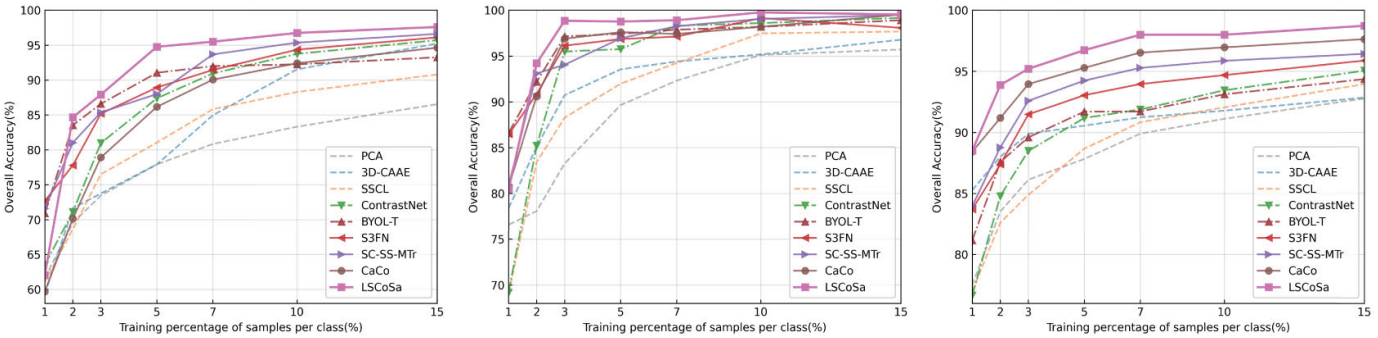


Fig. 9. Results of OA (%) as a function of percent training samples per class: IP (left), BO (middle), and HR-D (right).

are employed for classifier training, our LSCoSa can quickly recover to a stable and significant advantage. In particular, it consistently shows outstanding performance on the HR-D dataset.

V. Conclusion

This article presents an LSCoSa model for discriminative feature learning. Facing severe spectral uncertainty, such as intraclass diversity and interclass similarity of HSI, LSCoSa implements an online training strategy where positives and negatives are iteratively updated toward query anchors. Meanwhile, a KL divergence regularizer is adopted to suppress the prototype sampling, by which the learned dictionary can trace density distribution better on representations. Furthermore, multiple positive sample learning is proposed to cover more satisfying semantic variations. Extensive experiments show that our LSCoSa improves considerably over previous methods for self-supervised representation pretraining in HSIC.

CL with learnable sparse sampling can push away negatives and attempt to preserve intraclass diversity but cannot guarantee data representation with hierarchical semantic structures. Therefore, our future work will focus on enhancing pretraining modeling on downstream tasks with the hierarchical semantic representation of data.

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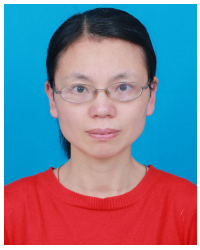
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